

A Journal of the Gesellschaft Deutscher Chemiker

Angewandte Chemie

GDCh

International Edition

www.angewandte.org

Accepted Article

Title: Stable Anion-Derived Solid Electrolyte Interphase by Regulating the Electrolyte Structure of Anions in Lithium Metal Batteries

Authors: Tao Li, Xue-Qiang Zhang, Nan Yao, Yu-Xing Yao, Li-Peng Hou, Xiang Chen, Ming-Yue Zhou, Jia-Qi Huang, and Qiang Zhang

This manuscript has been accepted after peer review and appears as an Accepted Article online prior to editing, proofing, and formal publication of the final Version of Record (VoR). This work is currently citable by using the Digital Object Identifier (DOI) given below. The VoR will be published online in Early View as soon as possible and may be different to this Accepted Article as a result of editing. Readers should obtain the VoR from the journal website shown below when it is published to ensure accuracy of information. The authors are responsible for the content of this Accepted Article.

To be cited as: *Angew. Chem. Int. Ed.* 10.1002/anie.202107732

Link to VoR: <https://doi.org/10.1002/anie.202107732>

COMMUNICATION

Stable Anion-Derived Solid Electrolyte Interphase in Lithium Metal Batteries

Tao Li[†], Xue-Qiang Zhang[†], Nan Yao, Yu-Xing Yao, Li-Peng Hou, Xiang Chen, Ming-Yue Zhou, Jia-Qi Huang, and Qiang Zhang^{*}

- [a] Dr. T. Li, N. Yao, Y.-X. Yao, L.-P. Hou, Dr. X. Chen, Dr. M.-Y. Zhou, Prof. Q. Zhang
Beijing Key Laboratory of Green Chemical Reaction Engineering and Technology
Department of Chemical Engineering
Tsinghua University
Beijing 100084 (P.R. China)
E-mail: zhang-qiang@mails.tsinghua.edu.cn
- [b] Dr. X.-Q. Zhang, Prof. J.-Q. Huang
Advanced Research Institute of Multidisciplinary Science
Beijing Institute of Technology
Beijing 100081 (P.R. China)
- [c] Dr. T. Li
School of Resource Environment and Safety Engineering
University of South China
Hengyang 421001 (P.R. China)
- [d] Dr. X.-Q. Zhang
Shanxi Research Institute for Clean Energy
Tsinghua University
Taiyuan 030032, Shanxi (P.R. China)
- [+][†] These authors contributed equally to this work.

Supporting information for this article is given via a link at the end of the document.

Abstract: High-energy-density lithium (Li) metal batteries are severely hindered by the dendritic Li deposition dictated by non-uniform solid electrolyte interphase (SEI). Despite its unique advantages in improving the uniformity of Li deposition, the current anion-derived SEI is unsatisfactory under practical conditions. Herein regulating the electrolyte structure of anions by anion receptors was proposed to construct stable anion-derived SEI. Tris(pentafluorophenyl)borane (TPFPB) anion acceptors with electron-deficient boron atoms interact with bis(fluorosulfonyl)imide anions (FSI⁻) and decrease the reduction stability of FSI⁻. Furthermore, the type of aggregate cluster of FSI⁻ in electrolyte changes, FSI⁻ interacting with more Li ions in the presence of TPFPB. Therefore, the decomposition of FSI⁻ to form Li₂S is promoted, improving the stability of anion-derived SEI. In working Li | LiNi_{0.5}Co_{0.2}Mn_{0.3}O₂ batteries under practical conditions, the anion-derived SEI with TPFPB undergoes 194 cycles compared with 98 cycles of routine anion-derived SEI. This work inspires a fresh ground to construct stable anion-derived SEI by manipulating the electrolyte structure of anions.

High-energy-density and long-cycling rechargeable batteries are necessary for the pursuit of decarbonization and wireless society. Lithium (Li) metal batteries are revived due to the high practical specific energy density over 350 Wh kg⁻¹.^[1] Nevertheless, practical Li metal batteries are severely hindered by the dendritic Li deposition which leads to a short cycling life and even safety risks.^[2] In common wisdom, dendritic Li deposition is induced by the heterogeneous and non-uniform solid electrolyte interphase (SEI) formed on Li metal anodes.^[3] Therefore, constructing stable and uniform SEI on Li metal anodes is necessary for practical Li metal batteries.

SEI is composed of the reduction decomposition products of electrolytes.^[4] The components and structure of SEI are mainly governed by electrolyte structure, *i.e.*, the microscopic interaction among solvents, anions, and Li ions.^[5] The electrolyte structure varies with not only the type of solvents and Li salts but also the

salt concentration.^[6] Recently, highly concentrated electrolytes (HCE) and localized high-concentration electrolytes (LHCE) exhibit unique advantages in stabilizing Li metal anodes by forming stable SEI.^[7] The molar ratio between solvents and Li salts is low (< 2) and anions are recruited into the first solvation sheath of Li ions, forming contact ion pairs (CIP) and aggregates (AGG) in HCE or LHCE.^[8] The components of SEI are then regulated by anions in HCE and LHCE, known as the anion-derived SEI.^[9] Despite its attractive performance in stabilizing Li metal anodes, the current anion-derived SEI is deficient in response to the challenges from practical conditions including an ultrathin Li anode (< 50 μm), a low negative/positive ratio (< 3), and a lean electrolyte (< 3 g Ah⁻¹) to achieve a long-cycling practical Li metal battery with a specific energy density over 350 Wh kg⁻¹.^[10] Therefore, the stability and uniformity of anion-derived SEI should be further improved to overcome the challenges under practical conditions.

Anions in the forms of CIP and AGG are the main precursors of anion-derived SEI.^[11] Regulating the ratio among CIP (FSI⁻ coordinating to one Li ion), AGG-I (FSI⁻ coordinating to two Li ions), and AGG-II (FSI⁻ coordinating to more than two Li ions) in electrolytes by the salt concentration and the type of solvents and diluents is proved to have a vital effect on dictating the components and structure of anion-derived SEI.^[12] Generally, the electrolyte structure of anions is indirectly regulated by Li ions because solvent and diluent molecules cannot directly interact with anions owing to their weakly localized positive charge.^[13] Therefore, novel strategies are highly expected to regulate the electrolyte structure of anions by directly interacting with anion.

In this contribution, an anion receptor, tris(pentafluorophenyl)borane (TPFPB), was proposed to directly regulate the electrolyte structure of anion via the electron-deficient boron (B) atom center. Through the direct interaction with TPFPB, FSI⁻ in forms of AGG-II significantly increases. In addition, the reduction stability of FSI⁻ decreases, which can be attributed to the interactions between TPFPB and FSI⁻ according

to density functional theory (DFT) calculations. Therefore, the decomposition of FSI^- to form Li_2S is promoted. In working Li | $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ (NCM523) batteries under practical conditions, the anion-derived SEI regulated by TFPFB delivered 194 cycles based on the benchmark of 80% capacity retention compared with 98 cycles of routine anion-derived SEI.

HCE and LHCE are two commonly used electrolyte systems for constructing anion-derived SEI. Considering the practicability of LHCE, a typical LHCE composed of LiFSI-dimethyl carbonate (DMC)-bis(2,2,2-trifluoroethyl) ether (BTFE, 1.0:1.5:2.0, by mol) was selected to construct a representative anion-derived SEI and this system is denoted as Base-LHCE. A typical boron-based anion receptor, TFPFB, was chosen as an additive to regulate the electrolyte structure of FSI^- via the dipole-dipole interactions, towards the goal of modifying the anion-derived SEI. Concretely, 2.0 wt.% TFPFB (TFPFB:LiFSI = 1:50, by mol) was added to the Base-LHCE after the optimization. The TFPFB-containing electrolyte is denoted as TFPFB-LHCE.

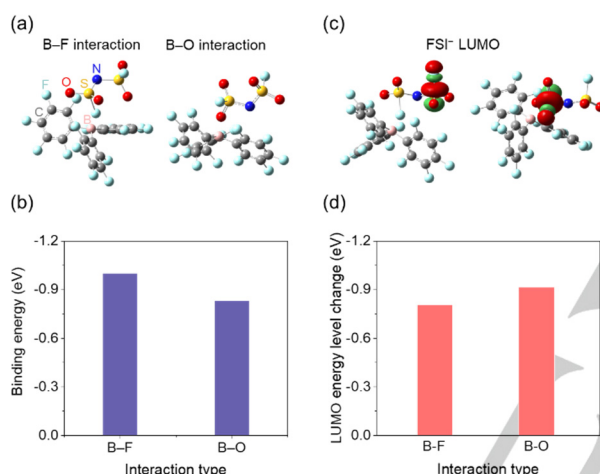


Figure 1. (a) The optimized geometrical structures of TFPFB- FSI^- through B-F or B-O interaction. (b) The binding energy of TFPFB- FSI^- through B-F or B-O interaction. (c) The visual LUMO and (d) LUMO energy level changes of FSI^- before and after interacting with TFPFB through B-F or B-O. The red and green regions of LUMOs represent the positive and negative parts of orbitals, respectively.

DFT calculations were performed to investigate the interaction between FSI^- and TFPFB. TFPFB molecules serve as anion receptors to attract FSI^- mainly through the interactions between the electron-deficient B atoms and F or O atoms (Figure 1a). The binding energy of TFPFB- FSI^- through B-F and B-O interaction is -1.00 and -0.83 eV (Figure 1b), respectively, demonstrating the relatively strong interaction between FSI^- and TFPFB. Furthermore, the lowest unoccupied molecular orbital (LUMO) energy level of FSI^- anion is reduced owing to B-F and B-O interactions (Figure 1c, d), indicating the reduction stability of FSI^- anions in TFPFB-LHCE is reduced. The linear sweep voltammetry (LSV) test was performed to compare the decomposition potential of FSI^- anions in Base-LHCE and TFPFB-LHCE (Figure S1). The peak assigned to the decomposition of FSI^- anions shifted from 1.39 to 1.45 V after adding TFPFB in Base-LHCE, verifying the reduced reduction stability of FSI^- anions.

Raman spectra were used to analyze the electrolyte structure of anions in HCE, Base-LHCE, and TFPFB-LHCE (Figure 2a and

S2). The vibration band of FSI^- at 710–780 cm^{-1} was deconvoluted into CIPs, AGG-I, and AGG-II.^[11] With the addition of BTFE in HCE, the peak area ratio assigned to AGG-II increases from 4% to 31% (Figure 2b), indicating that the introduction of diluents can strengthen the interaction between Li ions and FSI^- .^[12] In TFPFB-LHCE, the peak area ratio assigned to AGG-II further increases from 31% to 42% (Figure 2b), which is attributed to that the direct interaction between TFPFB and FSI^- reduces the electron density of this FSI^- anion and forces the originally solvated Li ions to interact with other FSI^- anions, forming new AGG-II. Based on DFT calculations and Raman spectra analysis, the effect of TFPFB on the electrolyte structure of FSI^- is summarized in a schematic diagram (Figure 2c, d). TFPFB can directly interact with FSI^- through B-F and B-O interactions, and decrease the reduction stability of FSI^- . In addition, the introduction of TFPFB increases the ratio of AGG-II in the electrolyte. The change of electrolyte structure of FSI^- is anticipated to have a significant impact on its decomposition behavior, and consequently altering the components of SEI.

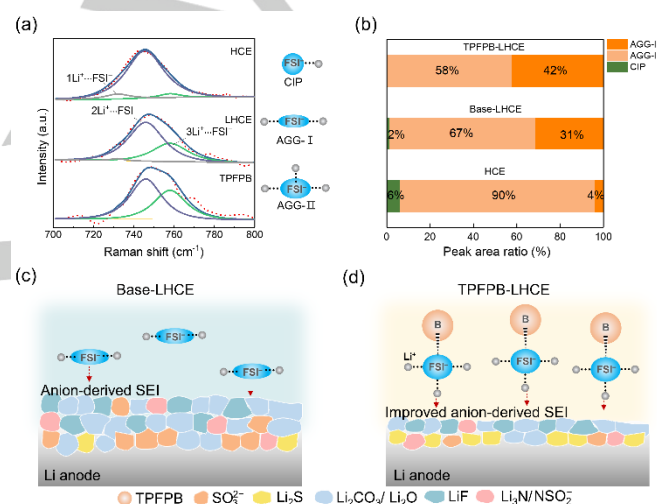


Figure 2. (a) Raman spectra of FSI^- in various aggregate clusters. (b) The peak area ratio of CIP, AGG-I, and AGG-II in HCE, Base-LHCE, and TFPFB-LHCE obtained from the fitting results of Raman spectra. Schematic diagram of the electrolyte structure of FSI^- and the correspondingly formed SEI in (c) Base-LHCE and (d) TFPFB-LHCE.

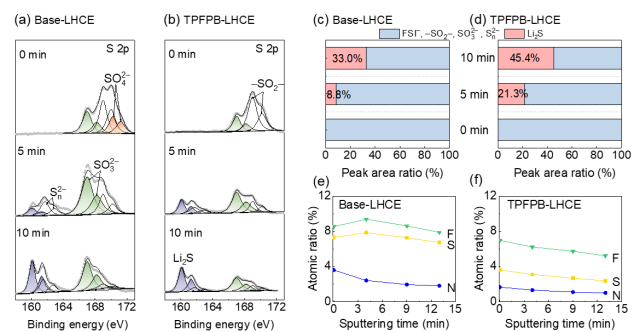


Figure 3. Component analysis of SEI formed in different electrolytes. XPS spectra of S 2p of the SEI formed in (a) Base-LHCE and (b) TFPFB-LHCE after 0, 5, and 10 min sputtering. The peak area ratio of different S-containing species in SEI formed in (c) Base-LHCE and (d) TFPFB-LHCE. The atomic ratio of F, N, and S in SEI formed in (e) Base-LHCE and (f) TFPFB-LHCE. Li metal anodes are retrieved from Li | NCM523 batteries in the charge state after 2 cycles.

X-ray photoelectron spectroscopy (XPS) was carried out to analyze the features of SEI formed in Base-LHCE and TFPB-LHCE. The components of anion-derived SEI are proven to be dramatically affected by the ratio among CIP, AGG-I, and AGG-II in electrolytes. Both SEIs featured an organic outer layer (high ratio of C-H/C-C) next to electrolytes and an inorganic inner layer (low C concentration, Figure S3 and S4) next to Li metal anode. S species with higher oxidation states ($\text{SO}_3^{2-}/\text{-SO}_2^-$) were situated on the outer layer of SEI, while more Li_2S can be detected closer to Li metal anode (Figure 3a, b). With the sputtering time increasing to 10 min, the striking difference in SEI was observed. A higher Li_2S ratio of 45.4% is detected in the SEI formed in TFPB-LHCE compared with 33.0% in the SEI formed in Base-LHCE (Figure 3c, d). Furthermore, the atomic ratios of F, S, and N in the SEI formed in TFPB-LHCE are lower than those in the SEI formed in Base-LHCE (Figure 3e, f and Figure S5), suggesting that less FSI^- decomposition occurs in TFPB-LHCE.

Based on the analysis of electrolyte structure and SEI components, the introduction of TFPB decreases the reduction stability of FSI^- and increases the amount of FSI^- in the forms of AGG-II. The decomposition behaviors of FSI^- are changed, resulting in the increased content of Li_2S in SEI, a complete decomposition product of FSI^- , which endows SEI with improved passivating ability.^[14]

The exchange current density (i_0) was measured to indirectly investigate the transport resistance of SEI formed in Base-LHCE and TFPB-LHCE. As shown in Figure S6, the exchange current density in TFPB-LHCE ($i_0=0.40 \text{ mA cm}^{-2}$) is almost twice that in Base-LHCE ($i_0=0.22 \text{ mA cm}^{-2}$). The increased exchange current density implies that the SEI formed in TFPB-LHCE renders a smaller transport resistance.

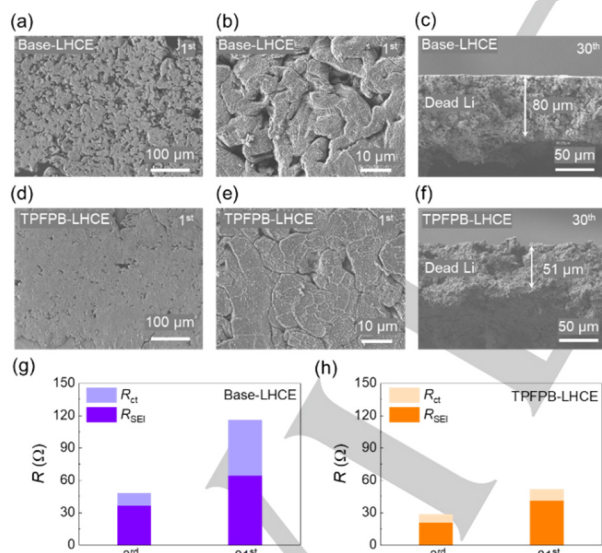


Figure 4. Bird and cross-sectional views of Li deposition morphology in Li | NCM523 batteries in (a–c) Base-LHCE and (d–f) TFPB-LHCE. (g–h) The evolution of interfacial resistance of Li | NCM523 batteries after 3 and 81 cycles in (g) Base-LHCE and (h) TFPB-LHCE.

The morphology of Li deposition can be regarded as a critical indicator for the uniformity of SEI. Li deposition morphology in Li | NCM523 batteries with different electrolytes after the initial charging process was characterized by scanning electron microscopy (SEM). Striking differences in the morphology of

deposited Li in different electrolytes were observed. Porous and loose Li deposition with nodule-like ($\sim 8 \mu\text{m}$ in diameter) morphology was observed in Base-LHCE (Figure 4a, b). In contrast, compact Li deposition with a larger diameter ($\sim 15 \mu\text{m}$ in diameter) was formed in TFPB-LHCE (Figure 4d, e). Compact and large Li deposition with a low surface area can greatly mitigate the parasitic reactions between Li and electrolytes. Therefore, the thickness of dead Li formed in TFPB-LHCE ($51 \mu\text{m}$) after 30 cycles is much thinner than that formed in Base-LHCE ($80 \mu\text{m}$, Figure 4c, f). The accumulation of dead Li induces the rise of interfacial resistance. The interfacial resistance evolution of Li | NCM523 batteries with different electrolytes was investigated by electrochemical impedance spectroscopy (EIS, Figure 4g, h, and S7), where the depressed semicircle at the high-frequency region represents the resistance of SEI (R_{SEI}) and the semicircle at middle-frequency region is assigned to the resistance of charge transfer (R_{ct}). The total interfacial resistance (R_{int}), that is the sum of R_{SEI} and R_{ct} , increases by 143% from the 3rd to 81st cycle in Base-LHCE (Table S1). In contrast, the increase of R_{int} in TFPB-LHCE is much smaller (82%). The results of EIS combined with Li deposition morphology indicate that the SEI formed in TFPB-LHCE improves the uniformity of Li ion transport in SEI and enables the compact Li deposition, thus mitigating the accumulation of dead Li and significantly decreasing the interfacial resistance.

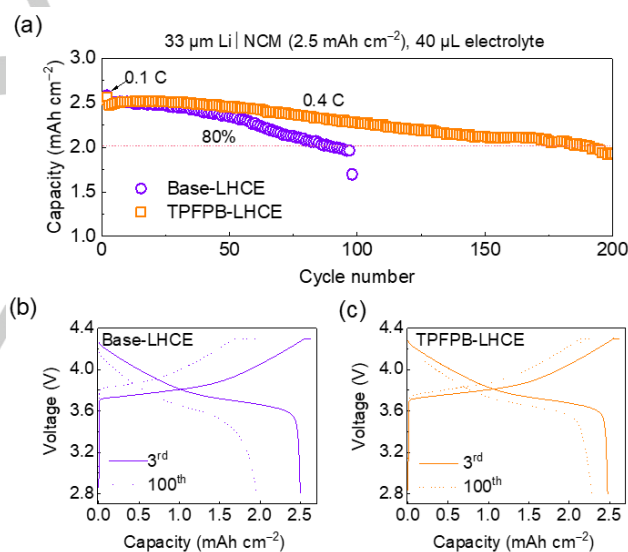


Figure 5. Electrochemical performance of Li | NCM523 batteries with different electrolytes. (a) Cycling performance at 0.4 C after two formation cycles at 0.1 C. The red dash line indicates 80% capacity retention of the initial discharge capacity at 0.4 C. Corresponding voltage profiles at the 3rd and 100th cycle in (b) Base-LHCE and (c) TFPB-LHCE.

The stability of SEI was evaluated in Li | NCM523 batteries under practical conditions, including a thin Li anode ($33 \mu\text{m}$), a high-loading NCM523 (2.5 mAh cm^{-2}) cathode, and a lean electrolyte ($40 \mu\text{L}$). As shown in Figure 5a, the discharge capacity of Li | NCM523 batteries cycled in Base-LHCE decays rapidly after 98 cycles. In contrast, Li | NCM523 batteries cycled in TFPB-LHCE maintain a capacity retention of 80% after 194 cycles. Correspondingly, TFPB-LHCE exhibits the slow increase of cell overpotentials during cycling (Figure 5b, c). The improved cycling performance of Li | NCM523 batteries under practical conditions verifies that the stability of anion-derived SEI is

significantly improved by regulating the electrolyte structure of anions *via* anion receptors.

In conclusion, stable anion-derived SEI was constructed by regulating the electrolyte structure of FSI⁻ anions *via* TPFPB anion acceptor. TPFPB can directly interact with FSI⁻ by electron-deficient B atoms. The decreased reduction stability of FSI⁻ anions was proved by DFT calculations and LSV measurements. In addition, the electrolyte structure of FSI⁻ anions, *i.e.*, the type of aggregate clusters, is significantly altered in presence of TPFPB. The ratio of AGG-II in electrolyte increases by 11%. Consequently, the decomposition behaviors of FSI⁻ anions change, affording to the increased content of Li₂S in SEI and the improved uniformity and stability of SEI. In Li | NCM523 batteries under practical conditions, the anion-derived SEI with TPFPB delivers 194 cycles in comparison to 98 cycles in common anion-derived SEI based on 80% retention. This work provides a fresh route to improve the stability of anion-derived SEI by manipulating the electrolyte structure of anions in electrolytes.

Acknowledgements

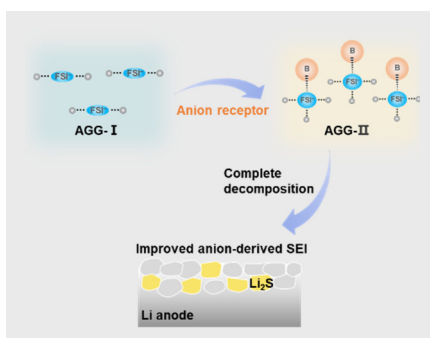
This work was supported by Beijing Municipal Natural Science Foundation (Z20J00043), National Natural Science Foundation of China (22061132002 and 21825501), the Seed Fund of Shanxi Research Institute for Clean Energy (SXKYJF015), and Beijing Natural Science Foundation (JQ20004, L182021).

Keywords: anion-derived solid electrolyte interphase • anion receptor • tris(pentafluorophenyl) borane • electrolyte structure • lithium metal batteries

- [1] a) X. B. Cheng, R. Zhang, C. Z. Zhao, Q. Zhang, *Chem. Rev.* **2017**, *117*, 10403–10473; b) J. Liu, Z. Bao, Y. Cui, E. J. Dufek, J. B. Goodenough, P. Khalifah, Q. Li, B. Y. Liaw, P. Liu, A. Manthiram, Y. S. Meng, V. R. Subramanian, M. F. Toney, V. V. Viswanathan, M. S. Whittingham, J. Xiao, W. Xu, J. Yang, X.-Q. Zhang, J.-G. Zhang, *Nat. Energy* **2019**, *4*, 180–186; c) X. Shen, X.-Q. Zhang, F. Ding, J.-Q. Huang, R. Xu, X. Chen, C. Yan, F.-Y. Su, C.-M. Chen, X. Liu, Q. Zhang, *Energy Mater. Adv.* **2021**, *2021*, 1205324.
- [2] a) J.-G. Zhang, W. Xu, J. Xiao, X. Cao, J. Liu, *Chem. Rev.* **2020**, *120*, 13312–13348; b) X. Zheng, L. Huang, X. Ye, J. Zhang, F. Min, W. Luo, Y. Huang, *Chem* **2021**, DOI: 10.1016/j.chempr.2021.02.025; c) X. Fan, L. Chen, O. Borodin, X. Ji, J. Chen, S. Hou, T. Deng, J. Zheng, C. Yang, S.-C. Liou, K. Amine, K. Xu, C. Wang, *Nat. Nanotechnol.* **2018**, *13*, 715–722; d) X. B. Cheng, C. Zhao, Y. Yao, H. Liu, Q. Zhang, *Chem* **2019**, *5*, 74–96; e) R. Xu, X. Cheng, C. Yan, X. Zhang, Y. Xiao, C. Zhao, J. Huang, Q. Zhang, *Matter* **2019**, *1*, 317–344.
- [3] a) H. Wu, H. Jia, C. Wang, J.-G. Zhang, W. Xu, *Adv. Energy Mater.* **2021**, *11*, 2003092; b) E. Peled, S. Menkin, *J. Electrochem. Soc.* **2017**, *164*, A1703–A1719; c) X.-R. Chen, C. Yan, J.-F. Ding, H.-J. Peng, Q. Zhang, *J. Energy Chem.* **2021**, *62*, 289–294; d) C. Yan, H. Yuan, H. S. Park, J.-Q. Huang, *J. Energy Chem.* **2020**, *47*, 217–220.
- [4] a) K. Xu, *Chem. Rev.* **2014**, *114*, 11503–11618; b) Y. Liu, D. Lin, Y. Li, G. Chen, A. Pei, O. Nix, Y. Li, Y. Cui, *Nat. Commun.* **2018**, *9*, 3656; c) L. Suo, W. Xue, M. Gobet, S. G. Greenbaum, C. Wang, Y. Chen, W. Yang, Y. Li, J. Li, *Proc. Natl. Acad. Sci. USA* **2018**, *115*, 1156–1161; d) X.-Q. Zhang, T. Li, B.-Q. Li, R. Zhang, P. Shi, C. Yan, J.-Q. Huang, Q. Zhang, *Angew. Chem. Int. Ed.* **2020**, *59*, 3252–3257; *Angew. Chem.* **2020**, *132*, 3278–3283.
- [5] a) H. Yang, J. Li, Z. Sun, R. Fang, D.-W. Wang, K. He, H.-M. Cheng, F. Li, *Energy Storage Mater.* **2020**, *30*, 113–129; b) S. S. Zhang, *InfoMat* **2021**, *3*, 125–130; c) J. Tan, J. Matz, P. Dong, J. Shen, M. Ye, *Adv. Energy Mater.* **2021**, *11*, 2100046; d) X. Liu, X. Shen, H. Li, P. Li, L. Luo, H. Fan, X. Feng, W. Chen, X. Ai, H. Yang, Y. Cao, *Adv. Energy Mater.* **2021**, *11*, 2003905.
- [6] a) J. Chen, T. Liu, L. Gao, Y. Qian, Y. Liu, X. Kong, *J. Energy Chem.* **2021**, *60*, 178–185; b) J. Qian, W. A. Henderson, W. Xu, P. Bhattacharya, M. Engelhard, O. Borodin, J.-G. Zhang, *Nat. Commun.* **2015**, *6*, 6362; c) Y.-S. Hu, Y. Lu, *ACS Energy Lett.* **2020**, *5*, 3633–3636; d) X.-Q. Zhang, X. Chen, X.-B. Cheng, B.-Q. Li, X. Shen, C. Yan, J.-Q. Huang, Q. Zhang, *Angew. Chem. Int. Ed.* **2018**, *57*, 5301–5305; *Angew. Chem.* **2018**, *130*, 5399–5403; e) K. Yoshida, M. Nakamura, Y. Kazue, N. Tachikawa, S. Tsuzuki, S. Seki, K. Dokko, M. Watanabe, *J. Am. Chem. Soc.* **2011**, *133*, 13121–13129.
- [7] a) S. Chen, J. Zheng, D. Mei, K. S. Han, M. H. Engelhard, W. Zhao, W. Xu, J. Liu, J.-G. Zhang, *Adv. Mater.* **2018**, *30*, 1706102; b) X. Ren, L. Zou, X. Cao, M. H. Engelhard, W. Liu, S. D. Burton, H. Lee, C. Niu, B. E. Matthews, Z. Zhu, C. Wang, B. W. Arey, J. Xiao, J. Liu, J.-G. Zhang, W. Xu, *Joule* **2019**, *3*, 1662–1676; c) X. Ren, S. Chen, H. Lee, D. Mei, M. H. Engelhard, S. D. Burton, W. Zhao, J. Zheng, Q. Li, M. S. Ding, M. Schroeder, J. Alvarado, K. Xu, Y. S. Meng, J. Liu, J.-G. Zhang, W. Xu, *Chem* **2018**, *4*, 1877–1892; d) L. Yu, S. Chen, H. Lee, L. Zhang, M. H. Engelhard, Q. Li, S. Jiao, J. Liu, W. Xu, J.-G. Zhang, *ACS Energy Lett.* **2018**, *3*, 2059–2067.
- [8] a) Y. Yamada, A. Yamada, *J. Electrochem. Soc.* **2015**, *162*, A2406–A2423; b) Y. Yamada, A. Yamada, *Chem. Lett.* **2017**, *46*, 1056–1064; c) M. Li, C. Wang, Z. Chen, K. Xu, J. Lu, *Chem. Rev.* **2020**, *120*, 6783–6819.
- [9] a) T. Li, X.-Q. Zhang, P. Shi, Q. Zhang, *Joule* **2019**, *3*, 2647–2661; b) X. Fan, L. Chen, X. Ji, T. Deng, S. Hou, J. Chen, J. Zheng, F. Wang, J. Jiang, K. Xu, C. Wang, *Chem* **2017**, *4*, 174–185; c) Y. Yamada, J. Wang, S. Ko, E. Watanabe, A. Yamada, *Nat. Energy* **2019**, *4*, 269–280; d) C. Wu, Y. Zhou, X. Zhu, M. Zhan, H. Yang, J. Qian, *Acta Phys. Chim. Sin.* **2021**, *37*, 2008044.
- [10] a) X.-M. Wang, P. Shi, X.-Q. Zhang, A.-B. Chen, Q. Zhang, *Acta Phys. Sin.* **2020**, *69*, 228501; b) P. Shi, X.-Q. Zhang, X. Shen, B.-Q. Li, R. Zhang, L.-P. Hou, Q. Zhang, *Adv. Funct. Mater.* **2021**, *31*, 2004189; c) C. Fang, X. Wang, Y. S. Meng, *Trends Mater.* **2019**, *1*, 152–158; d) X.-M. Wang, X.-Q. Zhang, P. Shi, L.-P. Hou, M.-Y. Zhou, A. Chen, Q. Zhang, *Mater. Chem. Front.* **2021**, *5*, 2791–2797; e) J. F. Ding, R. Xu, C. Yan, B. Li, H. Yuan, J. Huang, *J. Energy Chem.* **2021**, *59*, 306–319; f) R. H. Wang, W. S. Cui, F. Chu, F. X. Wu, *J. Energy Chem.* **2020**, *48*, 145–159.
- [11] a) J. Wang, Y. Yamada, K. Sodeyama, C. H. Chiang, Y. Tateyama, A. Yamada, *Nat. Commun.* **2016**, *7*, 12032; b) Y. Yamada, K. Furukawa, K. Sodeyama, K. Kikuchi, M. Yaegashi, Y. Tateyama, A. Yamada, *J. Am. Chem. Soc.* **2014**, *136*, 5039–5046.
- [12] a) J.-F. Ding, R. Xu, N. Yao, X. Chen, Y. Xiao, Y.-X. Yao, C. Yan, J. Xie, J.-Q. Huang, *Angew. Chem. Int. Ed.* **2021**, *60*, 11442–11447; *Angew. Chem.* **2021**, *133*, 11543–11548; b) N. Piao, X. Ji, H. Xu, X. Fan, L. Chen, S. Liu, M. N. Garaga, S. G. Greenbaum, L. Wang, C. Wang, X. He, *Adv. Energy Mater.* **2020**, *10*, 1903568.
- [13] a) X.-Q. Zhang, X. Chen, L.-P. Hou, B.-Q. Li, X.-B. Cheng, J.-Q. Huang, Q. Zhang, *ACS Energy Lett.* **2019**, *4*, 411–416; b) S. Li, Z. Luo, L. Li, J. Hu, G. Zou, H. Hou, X. Ji, *Energy Storage Mater.* **2020**, *32*, 306–319.
- [14] a) Z. Li, L. Huai, S. Li, M. Ma, K. Luo, Y. Zhao, D. Wang, X. Sun, Z. Peng, *Energy Storage Mater.* **2021**, *37*, 491–500; b) H. Chen, A. Pei, D. Lin, J. Xie, A. Yang, J. Xu, K. Lin, J. Wang, H. Wang, F. Shi, D. Boyle, Y. Cui, *Adv. Energy Mater.* **2019**, *9*, 1900858; c) W. Xue, Z. Shi, M. Huang, S. Feng, C. Wang, F. Wang, J. Lopez, B. Qiao, G. Xu, W. Zhang, Y. Dong, R. Gao, Y. Shao-Horn, J. A. Johnson, J. Li, *Energy Environ. Sci.* **2020**, *13*, 212–220.

COMMUNICATION

Directly regulating the electrolyte structure of anions by anion receptor was proposed to construct stable anion-derived SEI. The introduction of an anion receptor decreases the reduction stability of FSI⁻ and increases the amount of FSI⁻ in the forms of AGG-II. Therefore, the decomposition of FSI⁻ to form Li₂S is promoted, improving the stability of anion-derived SEI under practical conditions.



Tao Li[†], Xue-Qiang Zhang[†], Nan Yao, Yu-Xing Yao, Li-Peng Hou, Xiang Chen, Ming-Yue Zhou, Jia-Qi Huang, and Qiang Zhang^{*}

Page No. – Page No.

Stable Anion-Derived Solid Electrolyte Interphase in Lithium Metal Batteries